



UPLB IdSC-FaNS Knowledge Hub:

**A Knowledge Management System
for**

Food and Nutrition Security

Jose Gabriel F. Santos



Issues in Research

- **Dispersed Information:** Research is stored across various databases, institutions, and sources, making it challenging for stakeholders to access and interpret essential data.
- **Difficulty Identifying Trends:** Key patterns, emerging trends, and gaps in the research are hard to identify and interpret
- **Differences in Opinion:** Opinions are varied and decision-making is stalled or inconclusive



Solution?

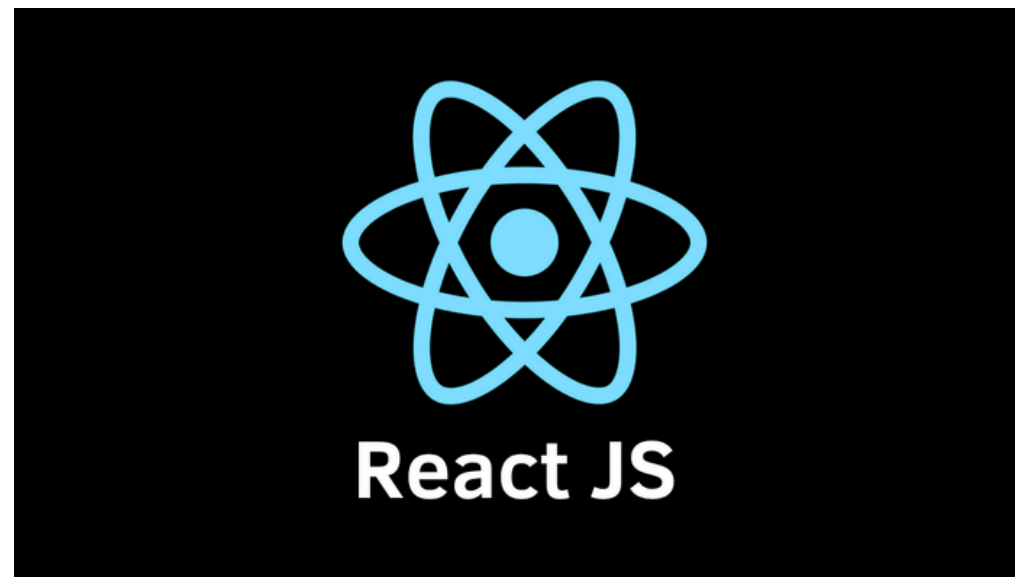
A Knowledge Management System!

Objectives

- 1) Provide a platform for research on food and nutrition security where the public can access curated and approved research.
- 2) Enable advanced search functionality for easy access to research data using Elasticsearch.
- 3) Provide data visualization and analytics tools that help users identify trends, emerging challenges, and research gaps in food security and nutrition.
- 4) Support community-driven curation through organizations and moderated discussions.
- 5) Provide AI-assisted curation through automated metadata extraction and research insight generation.



Tech Stack



NEXT.js



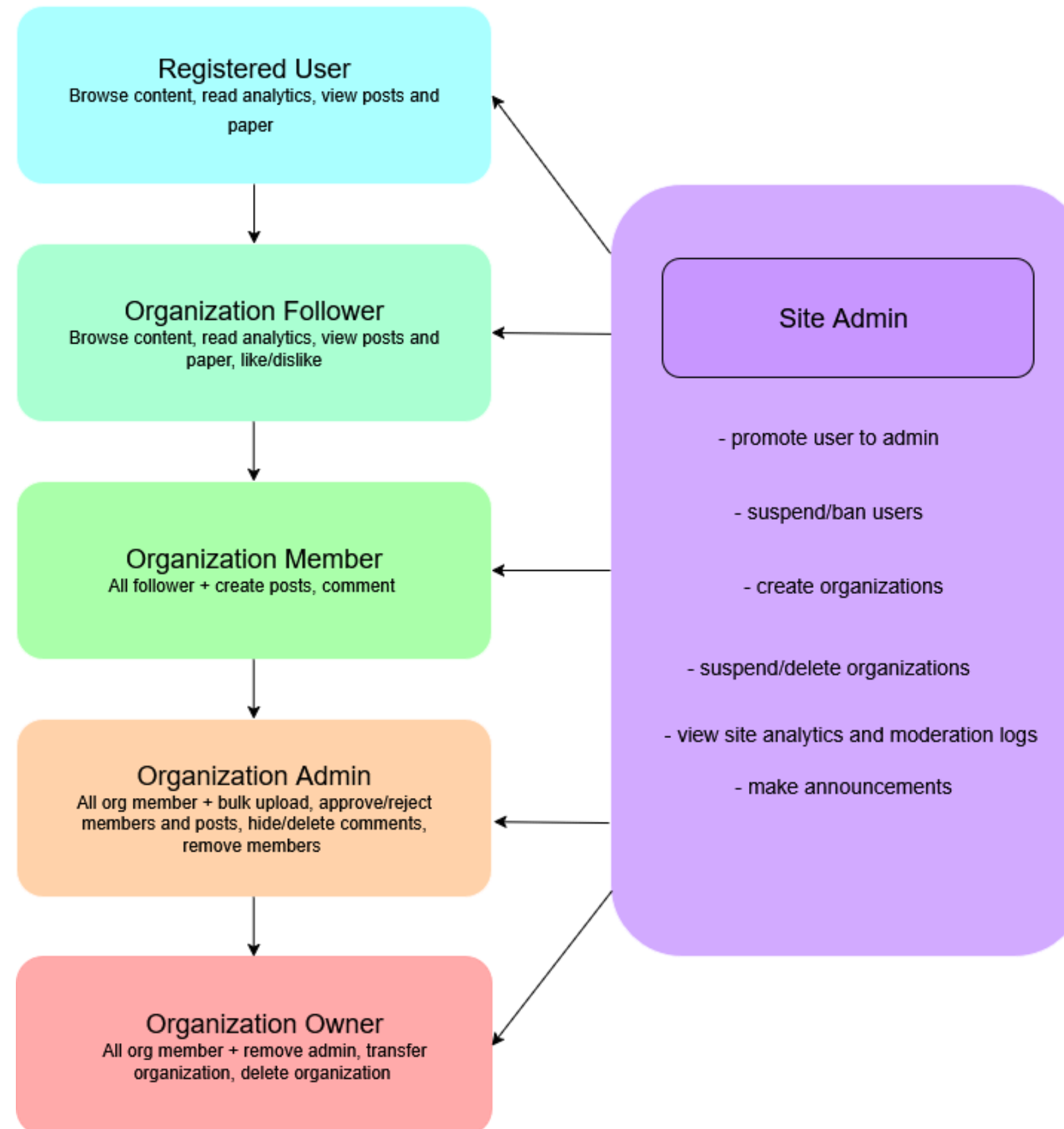
Tech Stack and Other Tools

- TailwindCSS
- TypeScript
- shadcn/ui
- TanStack Query
- Axios
- Socket.io
- Google OAuth
- Google Gemini 2.5 Flash



Gemini

Role-Based Access Control



Key Features



Centralized Repository



Advanced Search



Community
Engagements



Analytics and AI
Insights

Website Demo



UPLB FaNS Knowledge Hub

Food and Nutrition Security Research Platform

Log in

Sign up

UPLB Research Platform

UPLB FaNS Knowledge Hub

Food and Nutrition Security Research Platform

A community-driven platform for curated research on food and nutrition security. Discover, discuss, and visualize research insights — all in one place.

Get Started →

Log in to your account



uplb-fans-knowledge-hub.me/login



Testing and Evaluation

Instruments

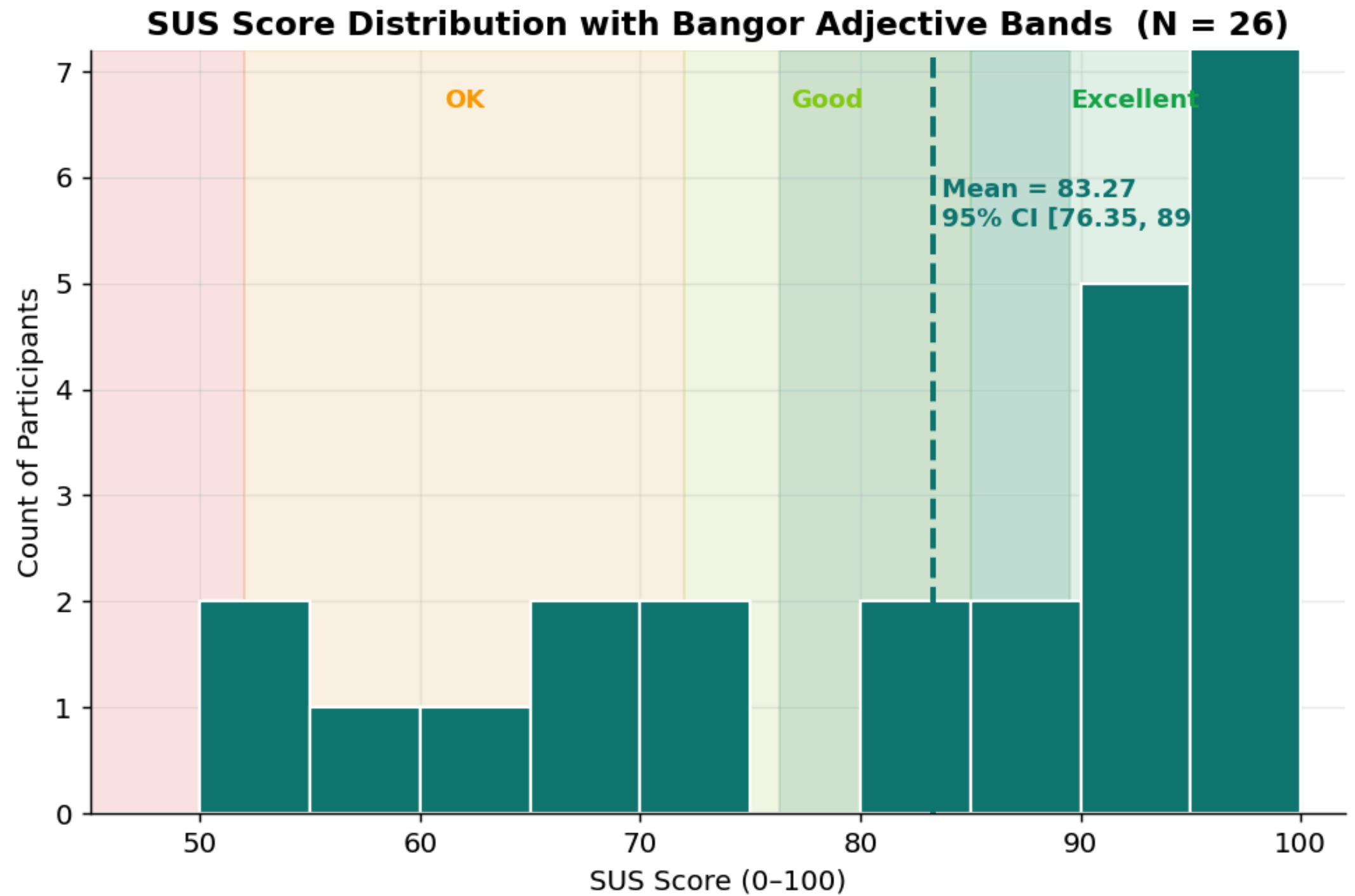
1. System Usability Scale
2. Task success rate
3. Single Ease Question
4. Feature-Specific Satisfaction
5. AI Insight Accuracy
6. AI Extraction Accuracy
7. Open-ended Feedback

Tracks and Participants

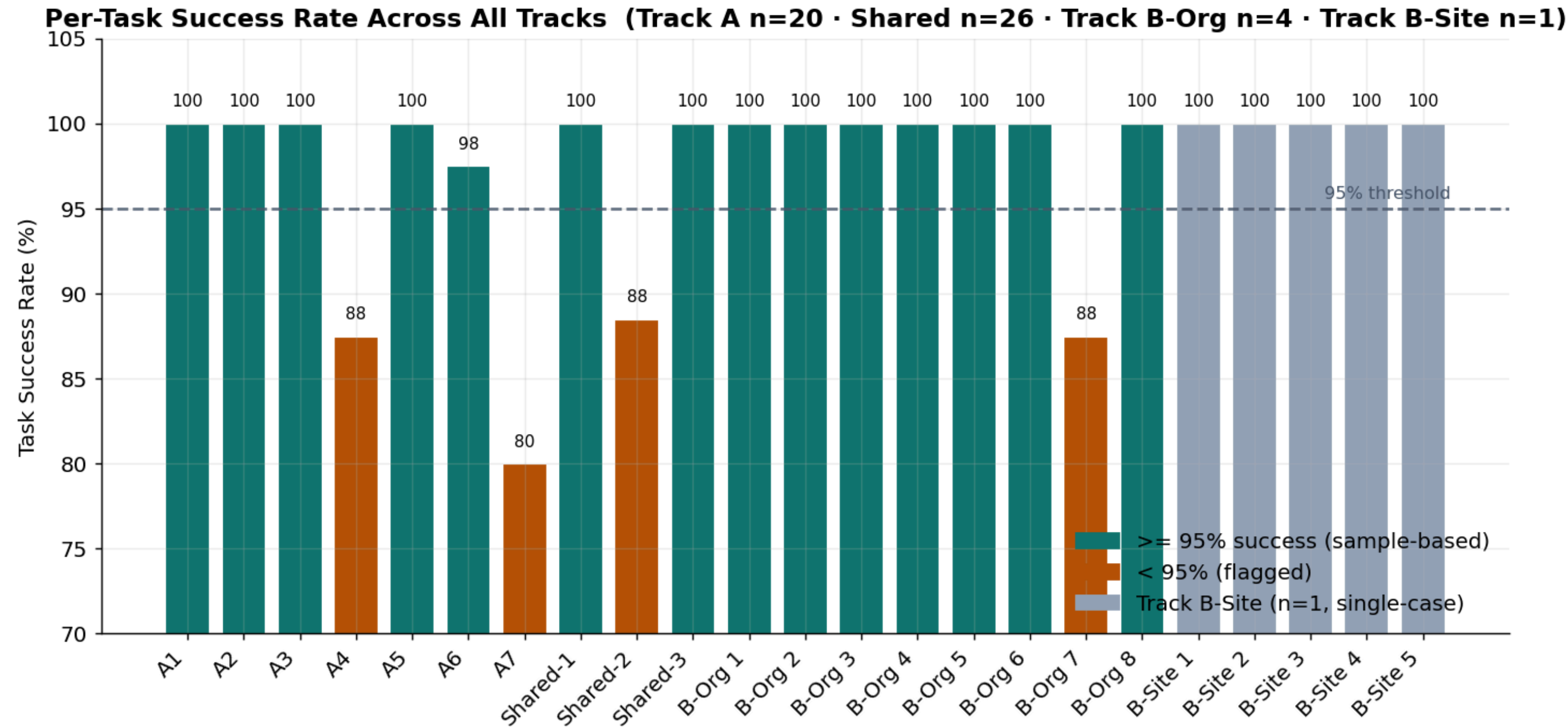
- **Track A - Standard User: 20**
- **Track B - Org: 5***
- **Track B - Site: 1**

System Usability Scale

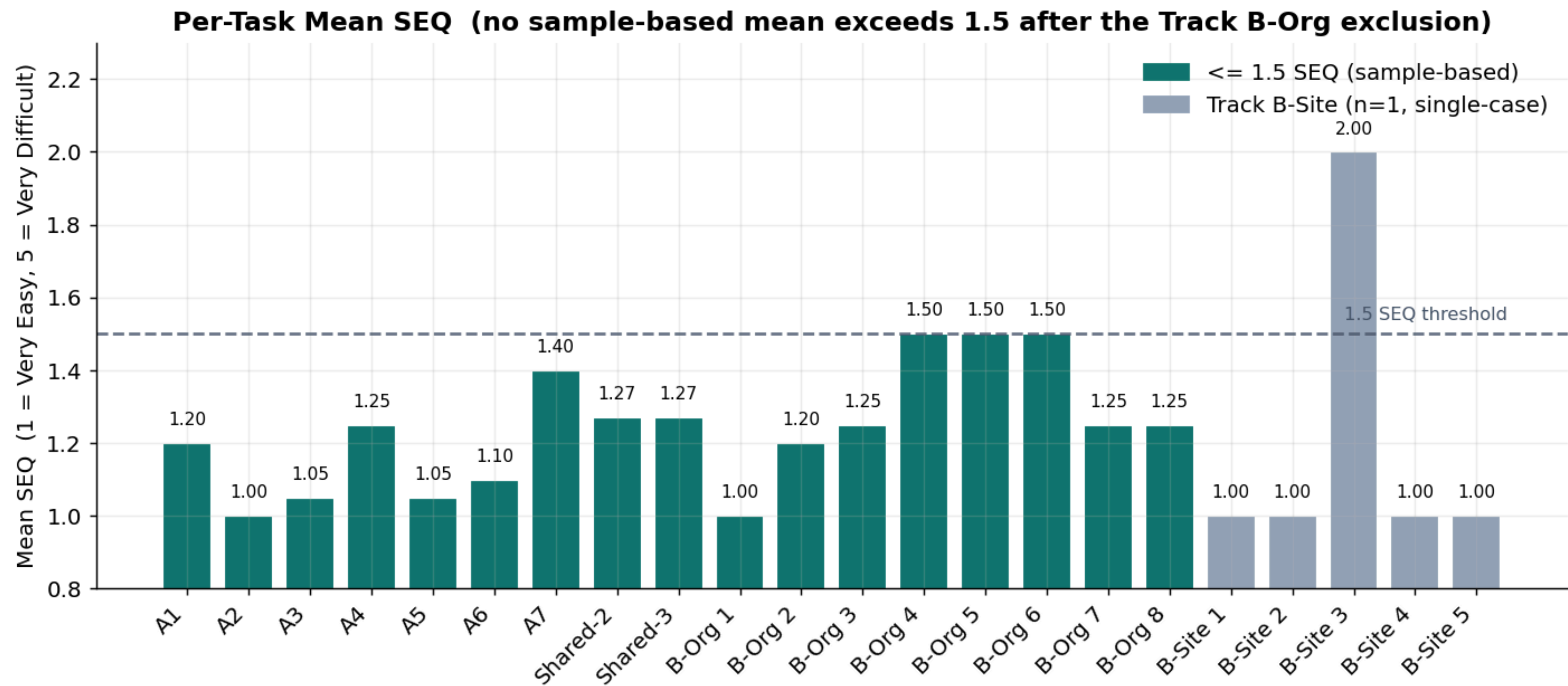
- mean 83.17
- median 92.5



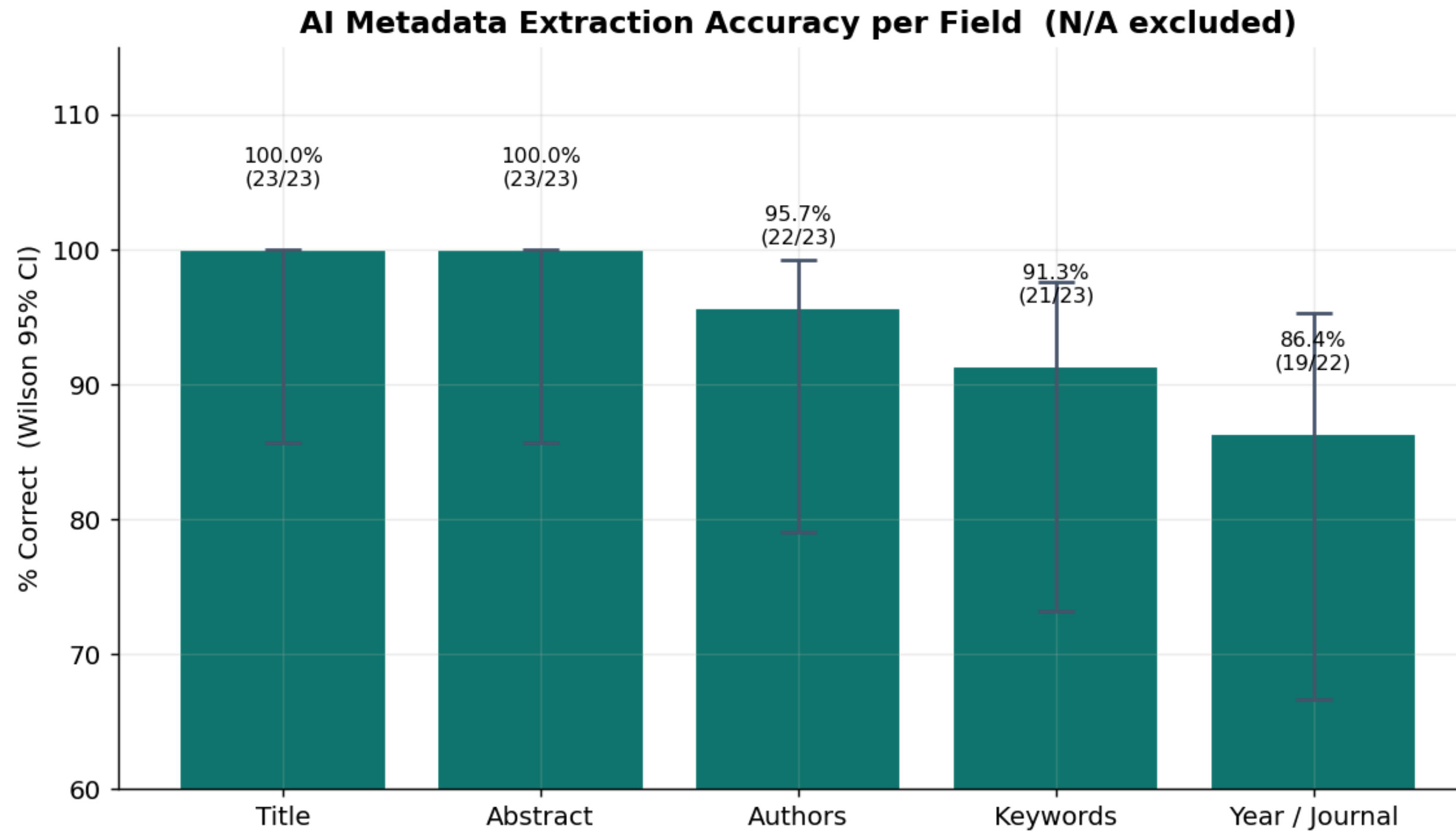
Task Success Rate



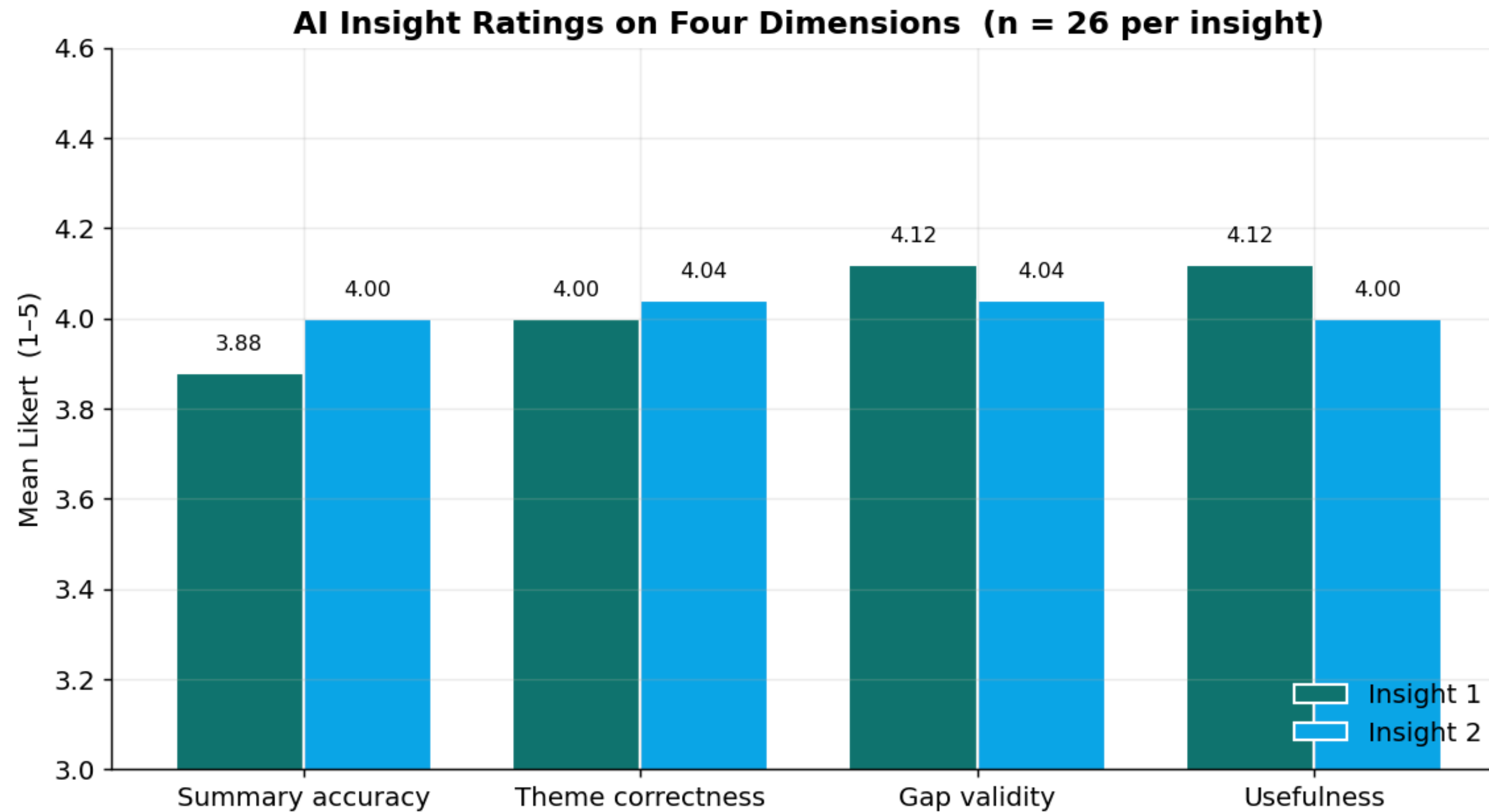
Single Ease Question



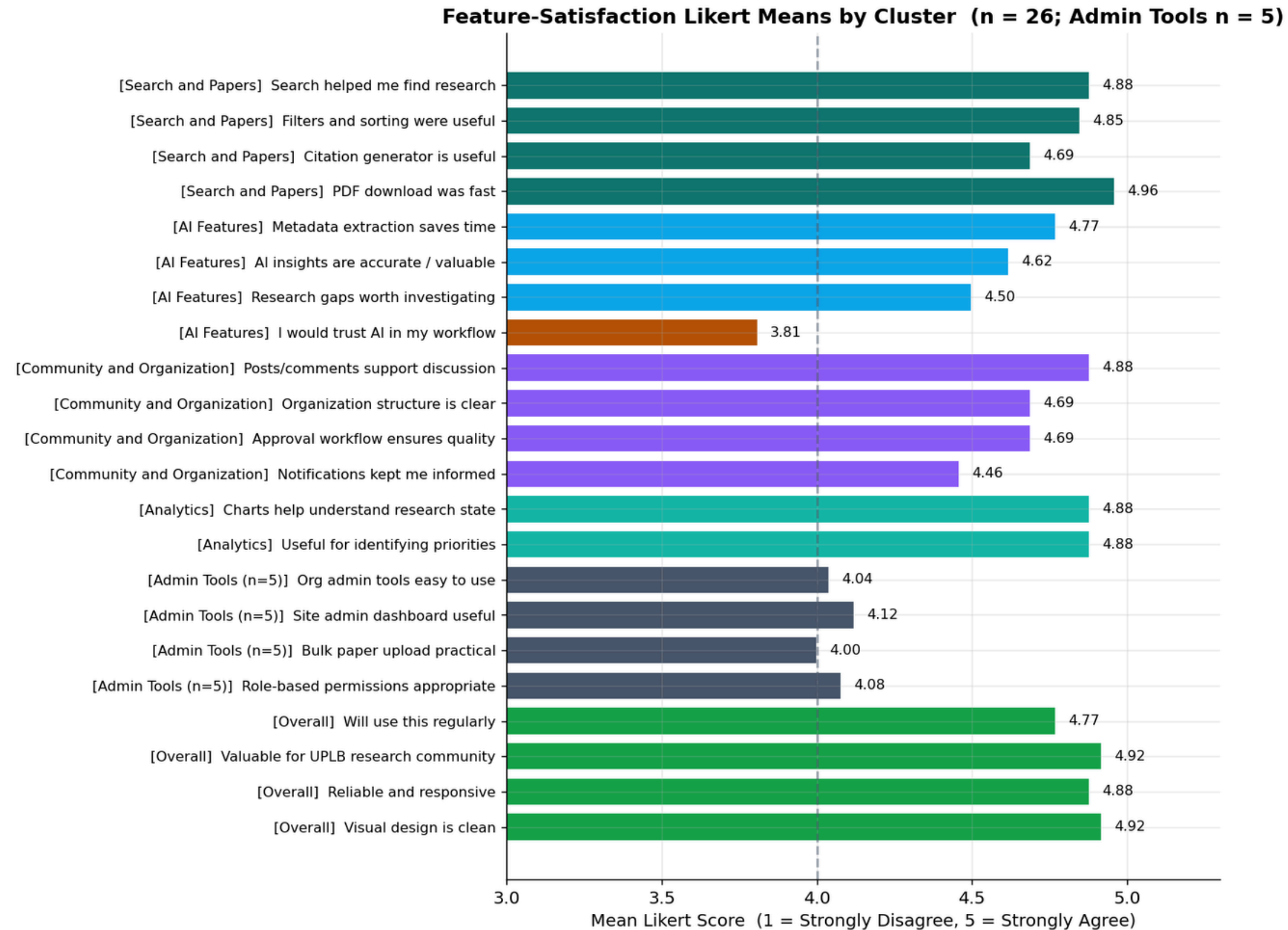
AI Accuracy (Extraction)



AI Accuracy (Insights)



Feature Satisfaction



Open-Ended Feedback

Most Useful Feature: Metadata Extraction

Most Confusing: Cite Feature (known bug)

Any Bugs?: internal server error on posting without a body

54.5% of users plan to explore the app further



Conclusion

The UPLB IdSC-FaNS Knowledge Hub meets all five research objectives at both the satisfaction and task-success layers, with a pooled SUS of 83.17 (Good–Excellent), 22 of 23 tasks at $\geq 95\%$ success, and AI metadata extraction named the most useful feature, leaving a focused bug-fix list rather than a structural rework.

It was able to provide a community-driven platform where research can be centralized and provide a more in-depth view of the food and nutrition research landscape through analytical and AI tools.

Limitations and Recommendation

The biggest limitations are the small admin samples: only 4 org admins (after dropping one whose account wasn't actually given admin permissions) and just 1 site admin, reported as a feasibility check rather than a real sample plus the fact that the unmoderated setup let that provisioning slip through unnoticed; the fix is a moderated follow-up that verifies roles up front, a short list of bug fixes, and pulling in more research sources beyond the initial UPLB Library dataset.

References

- [1] I. Angeles-Agdeppa et al., “Food sources, energy and nutrient intakes of adults: 2013 Philippines National Nutrition Survey,” Nutrition Journal, vol. 18, no. 1, Oct. 2019, doi: <https://doi.org/10.1186/s12937-019-0481-z>.
- [2] Francis Levi Durano, Jon Marx Sarmiento, G. Dee, G. D. Wahing, A. M. Traje, and D. Baker, “Food security of agri-food system actors during the COVID-19 pandemic in the Philippines: Post-pandemic implications to sustainable development,” Journal of Agriculture and Food Research, vol. 18, pp. 101284–101284, Dec. 2024, doi: <https://doi.org/10.1016/j.jafr.2024.101284>.
- [3] K. Heitman, J. Hubbard, L. Easter, and J. Kilkus, “Looking to the future: Agendas, directions, and resources for nutrition research,” Nutrition in clinical practice, Apr. 2024, doi: <https://doi.org/10.1002/ncp.11154>.
- [4] M. S. Righettini and E. Bordin, “Exploring food security as a multidimensional topic: twenty years of scientific publications and recent developments,” Quality & Quantity, Aug. 2022, doi: <https://doi.org/10.1007/s11135-022-01452-3>.
- [5] Global Nutrition Report, “Philippines Nutrition Profile,” Global Nutrition Report, Feb. 17, 2020. <https://globalnutritionreport.org/resources/nutrition-profiles/asia/south-eastern-asia/philippines/>